Nicholas Eisenberg, Ph.D.

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Experience

Senior Data Scientist, Nevada National Security Site — July 2022 - Present

- Managed a MySQL database of seismic sensor data using SQLAlchemy, guaranteeing data integrity and optimizing data retrieval for analysis purposes.
- Acted as the lead investigator for a high-impact project involving the organization and feature engineering of a dataset consisting of 10,000+ shots from a linear accelerator.
- Led predictive analytics efforts using TensorFlow to forecast future performance of an X-ray machine, resulting in improved maintenance planning and operational efficiency.
- Developed interactive dashboards utilizing Streamlit, Plotly, and Bokeh with Holoviews to visually present complex data and provide actionable insights to stakeholders.
- Applied Scikit-learn and Scikit-image to analyze and characterize X-ray machine spots, estimating the blur kernel for deblurring X-ray images, enhancing image quality, and aiding in accurate analysis.
- Maintained an internal Python library hosted on GitLab, ensuring code quality, version control, and collaboration among team members.

Graduate Teaching Assistant, Auburn University — August 2021 - June 2022

- Instructed a course in College Algebra
- Instructed weekly recitation sections for Mathematical Statistics
- Managed the math tutoring center

Graduate Teaching Assistant, University of Nevada Las Vegas — Janurary 2016 - Janurary 2018

- Instructed a courses in College Algebra and Pre-Calculus
- Instructed weekly recitation sections for Real Analysis and Calculus I, II and III

Technical Skills

Programming Languages: Python, SQL, R, Lua, Bash, LaTeX

Libraries: scikit-learn, tensorflow, keras, pytorch, scipy, plotly, steamlit, pymysql, sqlalchemy

Developer Tools: Git, Unix/Linix, Neovim, AWS

Education

Auburn University, Ph.D. — June 2022
Research Area: Probability (Stochastic Processes and Partial Differential Equtions)
Advisor: Le Chen

University of Nevada Las Vegas, B.S./M.S. — December 2015 / Janurary 2018
Major: Applied Mathematics

Publications

Chen, L., Eisenberg, N. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). https://doi.org/10.1007/s40072-022-00258-6

Chen, L., Eisenberg, N. Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term. Submitted Janurary 2023 to Journal of Theoretical Probability.
Preprint available at https://arxiv.org/pdf/2209.04771v1.pdf